ONLINE VOTING SYSTEM

Database Schema and System Diagrams

Prepared according to the supplied Software Requirements Specification (SRS)

Technology: HTML, CSS, JavaScript, PHP, MySQL and Apache (XAMPP).

Project scope: voter registration and login, viewing elections and candidates, casting one vote per election, administrator management and displaying results.

1. DATABASE SCHEMA

The database contains five main tables. Primary keys uniquely identify records, while foreign keys connect votes to voters, elections and candidates.

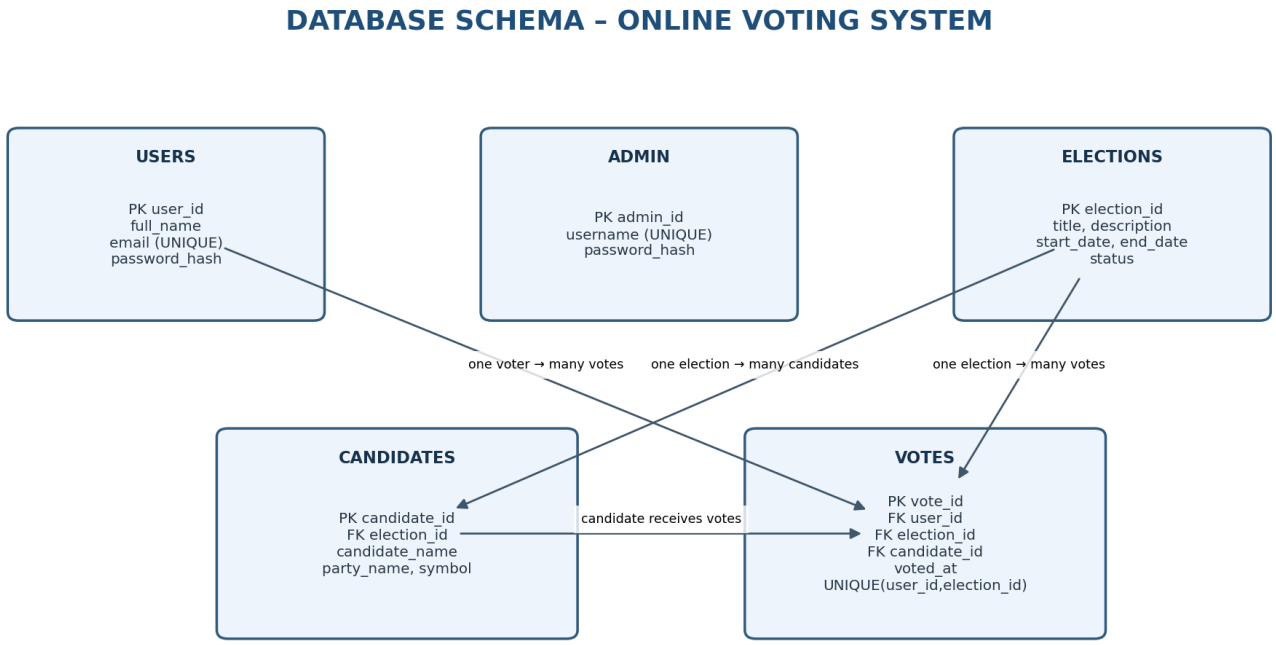


Table details

Table	Important fields	Purpose
users	user_id (PK), full_name, email (UNIQUE), password_hash, created_at	Stores registered voter details.
admin	admin_id (PK), username (UNIQUE), password_hash	Stores administrator login details.
elections	election_id (PK), title, description, start_date, end_date, status	Stores election information and status.
candidates	candidate_id (PK), election_id (FK), candidate_name, party_name, symbol	Stores candidates belonging to an election.
votes	vote_id (PK), user_id (FK), election_id (FK), candidate_id (FK), voted_at	Stores each submitted vote.

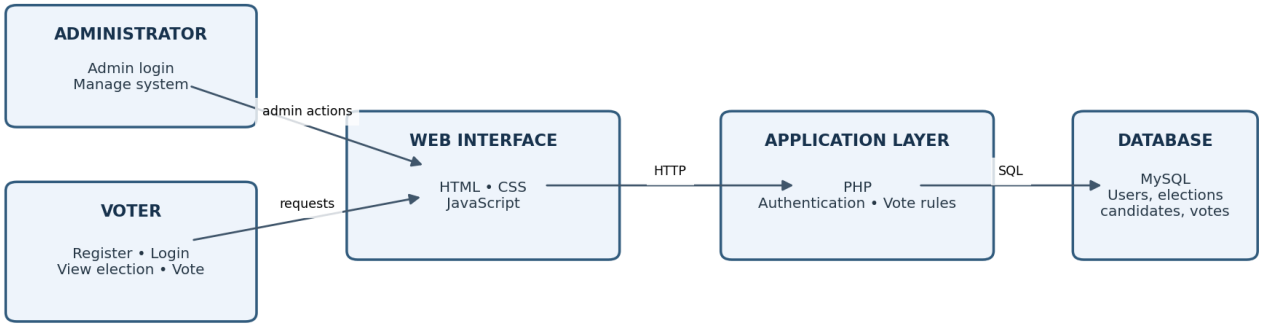
Important constraint: enforce UNIQUE(user_id, election_id) in the votes table. This prevents a voter from voting more than once in the same election.

Schema note: the SRS names the five tables but does not specify every column. The fields shown are a proposed implementation schema consistent with the stated requirements.

2. BLOCK DIAGRAM (BRD)

This diagram shows the main system components and how the voter and administrator interact with the web interface, PHP application layer and MySQL database.

BLOCK DIAGRAM (BRD) – ONLINE VOTING SYSTEM

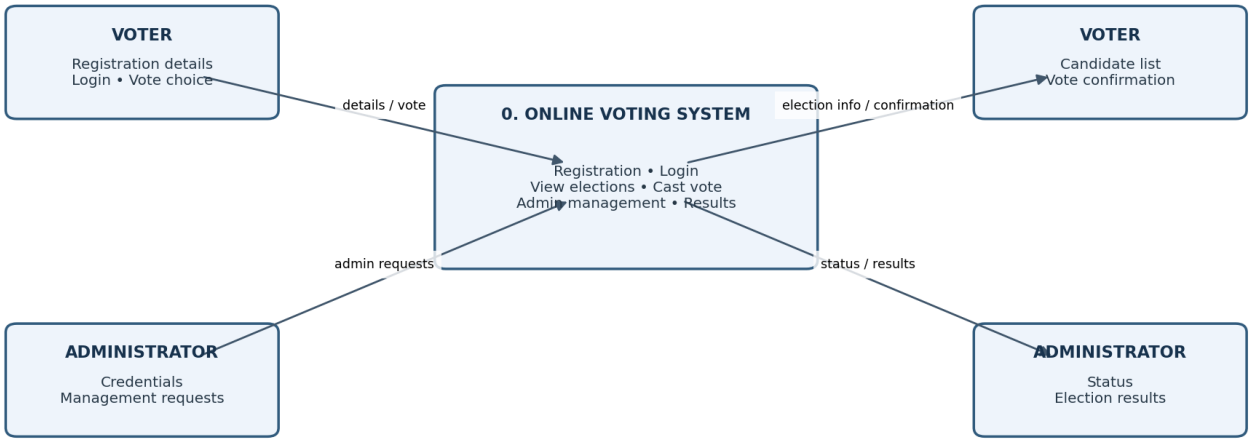


3. DATA FLOW DIAGRAM (DFD)

3.1 Level 0 – Context Diagram

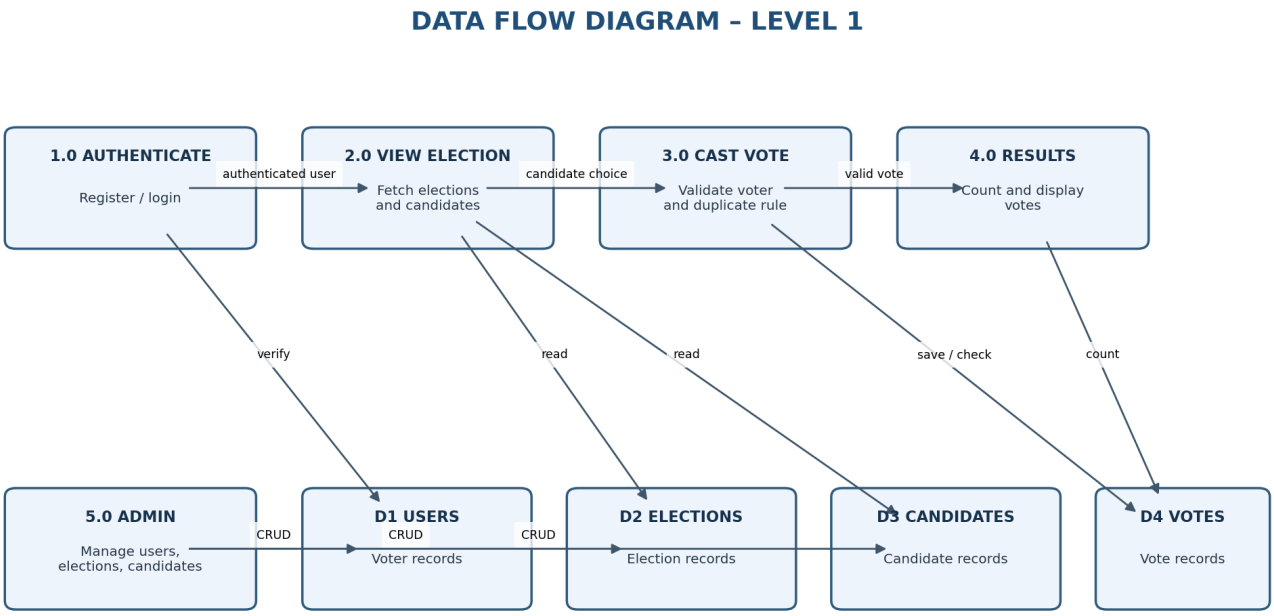
The complete Online Voting System is represented as one process. The voter and administrator provide input and receive the appropriate system outputs.

DATA FLOW DIAGRAM – LEVEL 0 (CONTEXT)



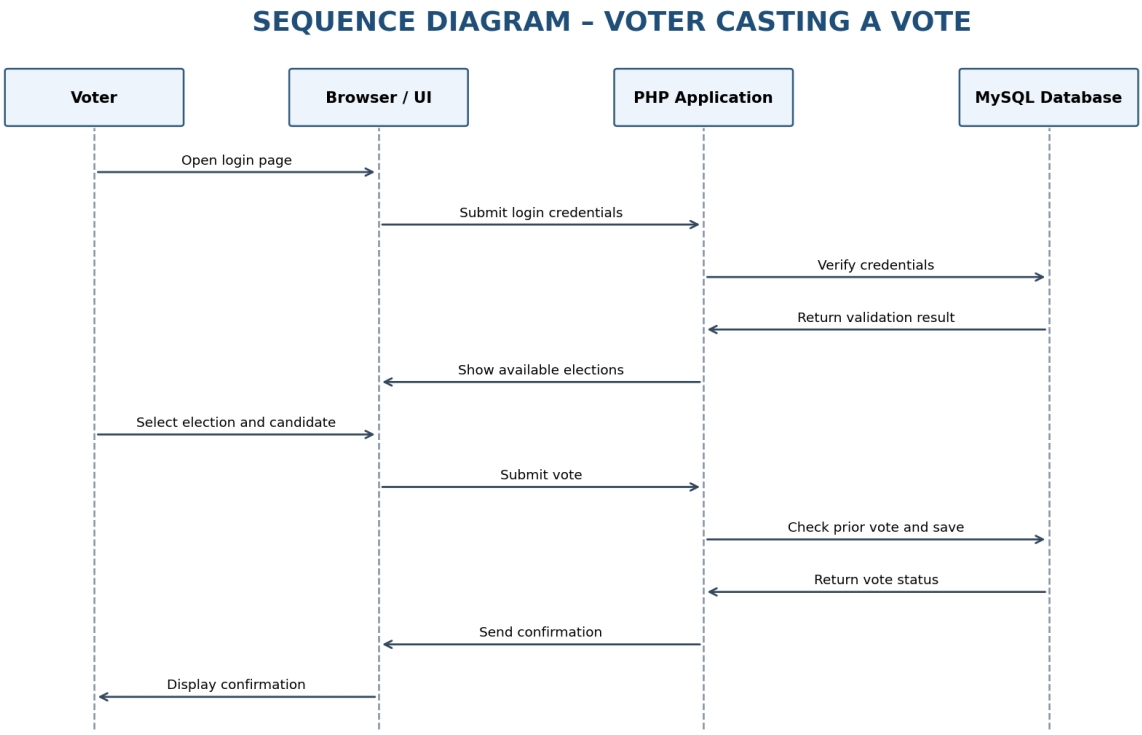
3.2 Level 1 – Detailed Data Flow

The system is divided into authentication, election viewing, vote casting, results and administrator-management processes. D1–D4 represent the database stores.



4. SEQUENCE DIAGRAM

The sequence diagram describes the normal voter journey: opening the login page, submitting credentials, viewing elections, selecting a candidate and submitting a vote. PHP checks the database before saving the vote and returning confirmation.



If the voter has already voted in this election, the application rejects the new submission.

5. IMPORTANT IMPLEMENTATION RULES

- Only registered voters can log in and vote.
- Only an authorized administrator can access the admin section.
- A voter can cast only one vote in each election.
- The selected candidate must belong to the election for which the vote is being submitted.
- Passwords should be stored as secure password hashes, not plain text.
- Vote insertion and duplicate-vote checking should be handled safely on the server side.

Prepared from the supplied Online Voting System SRS. Detailed database fields and diagram layouts are implementation-level interpretations where the SRS provides only high-level descriptions.